UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	PROGRAMMING I
COURSE CODE	:	CSC402
EXAMINATION	:	NOV 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC402 mock paper (November 2026). Generated answers show working; all questions are deterministic and paper-derived; no lecture slides are used.

2. Marking points show the expected answer and the allocation of marks; unresolved slots would remain visible.

3. Part A = 10 x 1 mark; Part B = 5 questions (60 marks); Part C = 2 x 15 marks; total = 100 marks.

4. Questions: 17. Slots: 22. Placeholders: 0. No LLM call. Questions and answers are for local study use only.

5. Structure derived from 21 unique CSC402 past papers; no past-paper text is copied.

FOR MARKERS' USE ONLY

This marking scheme consists of 6 printed pages

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PART A — ANSWER KEY

Q	Ans	Question	Solution
A1	D	Which item is NOT a programming paradigm?	Debugging is an activity, not a programming paradigm.
A2	A	How many times does a while loop execute when its condition is false initially?	while is a pretest loop; a false condition skips the body.
A3	D	Which identifier is valid in C++?	An identifier may contain letters, digits and underscores, but cannot start with a digit or be a keyword.
A4	A	Which statement immediately exits a loop?	break terminates the nearest loop or switch.
A5	B	Which declaration creates a floating-point variable named price?	float is used for real numbers.
A6	C	Which operator reads data from cin?	The extraction operator >> reads from an input stream.
A7	D	What does the % operator return?	% is the modulus operator and returns the remainder.
A8	C	Which part of a for loop is executed after each iteration?	The update expression runs after each iteration.
A9	A	What is pseudocode?	Pseudocode expresses algorithm logic without strict language syntax.
A10	C	Which loop is commonly used to process every array element?	A for loop with an index visits each element.

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PART B — WORKED ANSWERS

QUESTION 1

Topic: fundamentals

a) Define source code and object code.

Source code is the human-readable program text; object code is the machine-oriented output produced by a compiler.

Marking: Award 1 mark for each definition.

(3 marks)

b) Describe THREE stages of the program development life cycle.

Analysis (requirements), design (algorithm/flowchart), coding (implementation), testing (verification), maintenance (updates). Any three stages.

Marking: Award 1 mark for each correctly described stage.

(2 marks)

QUESTION 2

Topic: control structures

a) Draw a flowchart for the grade classification.

Start -> read score -> decision score >= 80 -> A; else score >= 60 -> B; else score >= 50 -> C; else score >= 40 -> D; else F -> display grade -> end.

Marking: Award marks for terminal symbols, input, four/five decisions, each output and correct flow.

(6 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

IF score >= 80 THEN grade = A ELSE IF score >= 60 THEN grade = B ... ENDIF; use if/else if in C++ and display grade.

Marking: Award marks for correct pseudocode, complete C++ structure, conditions, input and output.

(7 marks)

QUESTION 3

Topic: data types

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

Implicit conversion is automatic (int to double); explicit conversion uses a cast such as double(total) or static_cast<double>(total).

Marking: Award marks for both definitions and valid examples.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

double average = (m1 + m2 + m3) / 3.0; cout << fixed << setprecision(2) << average;

Marking: Award marks for floating-point division, fixed/setprecision and output.

(5 marks)

QUESTION 4

Topic: control structures

a) Trace the program and show the exact output.

1\n12\n123

Marking: Award marks for each line and the exact row lengths.

(10 marks)

b) Identify TWO logic errors in a version that prints four rows but reuses j without resetting it. Rewrite the corrected code.

The inner loop must initialise j = 1 for every row and the outer loop must run to 4. Correct code: for (int i=1;i<=4;i++){for(int j=1;j<=i;j++) cout<<j; cout<<endl;}

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Marking: Award marks for each error and the corrected loop.

(5 marks)

QUESTION 5

Topic: arrays

a) Declare an array named prices with the values 20.5, 55.0, 15.9, 75.0 and 10.0. Display all values.

double prices[5] = {20.5, 55.0, 15.9, 75.0, 10.0}; for (int i=0;i<5;i++) cout << prices[i] << endl;

Marking: Award marks for correct type, initialiser, loop and output.

(6 marks)

b) Write a program segment that applies a 10% discount to prices above RM50 and stores the results in a second array. Display both arrays.

double sale[5]; for(int i=0;i<5;i++){ sale[i] = prices[i] > 50 ? prices[i]*0.9 : prices[i]; } then display both arrays.

Marking: Award marks for second array, condition, discount, preservation and output.

(10 marks)

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PART C — WORKED ANSWERS

QUESTION 1

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

```
#include <iostream>
#include <string>
using namespace std;
struct Book {
    string title;
    double price;
    int quantity;
};
int main() {
    Book books[5];
    double totalValue = 0;
    int mostExpensive = 0;
    for (int i = 0; i < 5; i++) {
        cin.ignore();
        cout << "Title: ";
        getline(cin, books[i].title);
        cout << "Price and quantity: ";
        cin >> books[i].price >> books[i].quantity;
        totalValue += books[i].price * books[i].quantity;
        if (books[i].price > books[mostExpensive].price) {
            mostExpensive = i;
        }
    }
    cout << "Total inventory value: RM" << totalValue << endl;
    cout << "Most expensive: " << books[mostExpensive].title
         << " (RM" << books[mostExpensive].price << ")" << endl;
    return 0;
}
```

Marking: Award 3 marks for struct definition, 4 for input, 4 for inventory value, 3 for most expensive and 1 for output.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

```
#include <iostream>
using namespace std;
int main() {
    int marks[10], total = 0, highest, highestIndex = 0;
    int aboveAverage = 0;
    for (int i = 0; i < 10; i++) {
        cin >> marks[i];
        total += marks[i];
        if (i == 0 || marks[i] > highest) {
            highest = marks[i];
            highestIndex = i;
        }
    }
    double average = total / 10.0;
    for (int i = 0; i < 10; i++) {
        if (marks[i] > average) aboveAverage++;
    }
    cout << "Average: " << average << endl;
    cout << "Highest: " << highest
         << " at index " << highestIndex << endl;
    cout << "Above average: " << aboveAverage << endl;
    return 0;
}
```

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Marking: Award 3 marks for input, 4 for total/average, 3 for highest/index, 3 for above-average count and 2 for output.

(15 marks)

END OF MARKING SCHEME

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